



Place & Manner

Dr. Josef Fruehwald

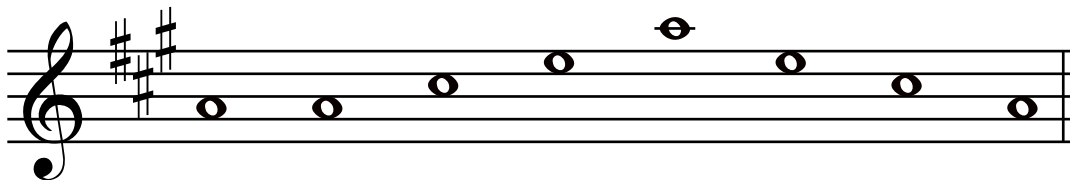
2026-09-03

Reading

Zsiga Chapter 2

Transcription

Transcription

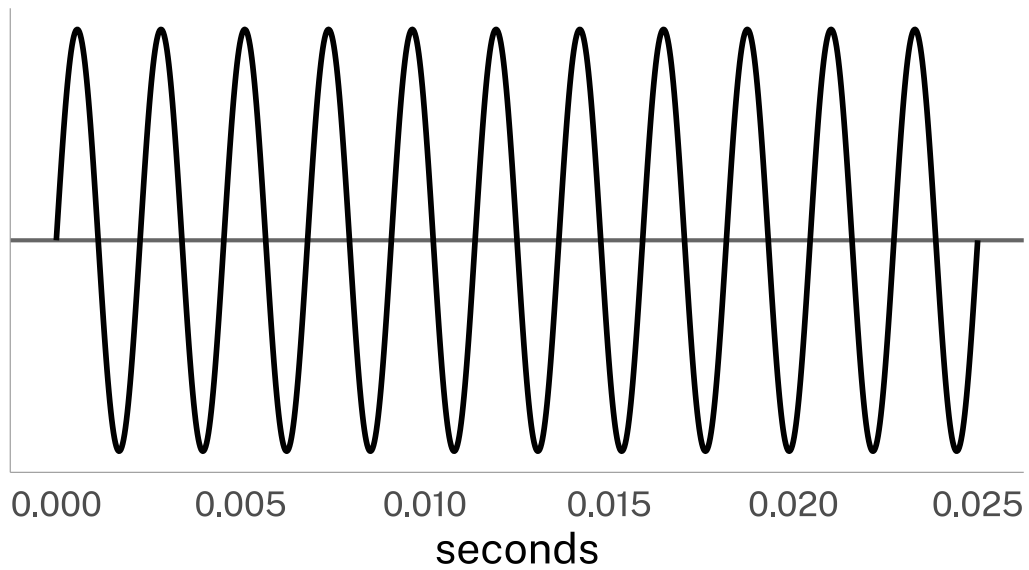


Notation

But what is “A”?



440 Hertz



Transcription

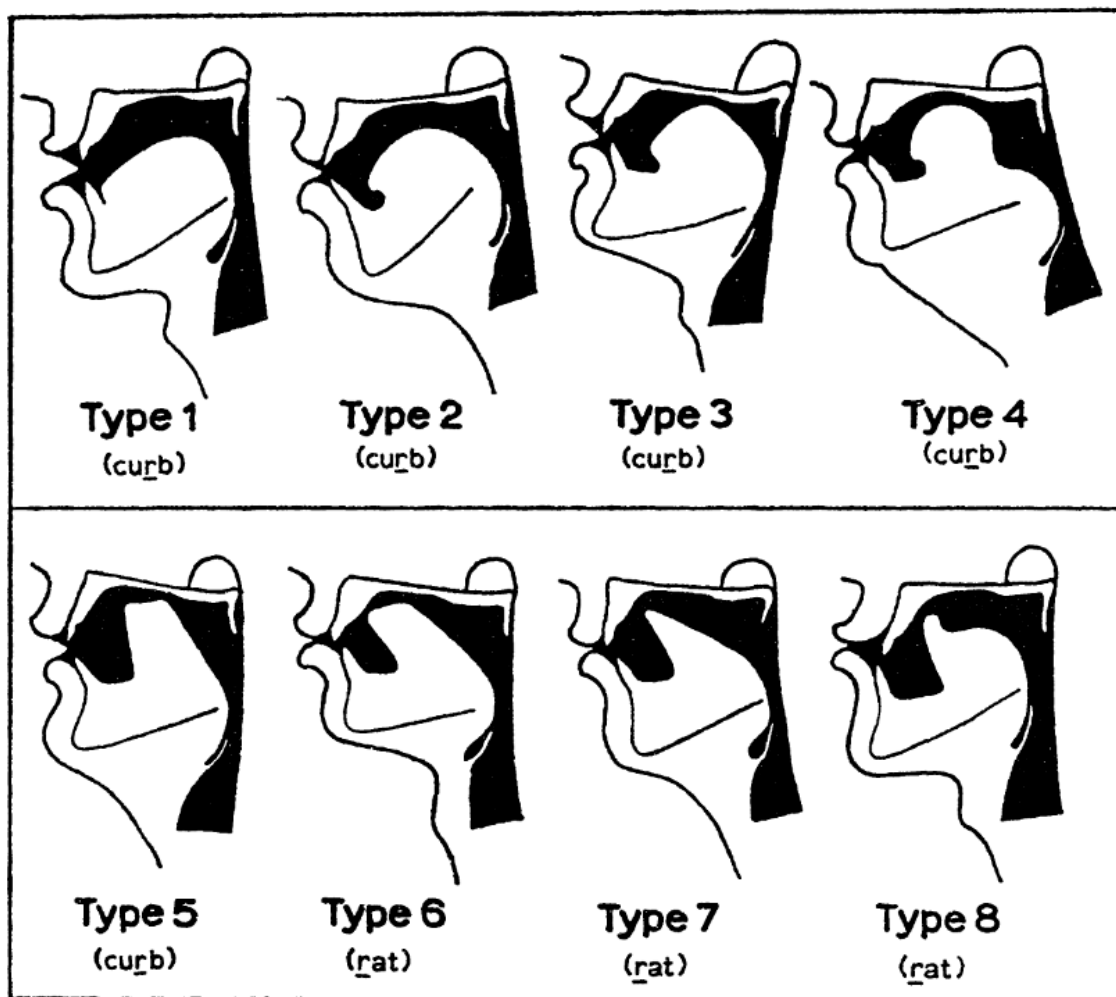
There have been different conventions for what frequency “Concert A” should be!

440Hz how?





ɪ how?



source: Delattre & Freeman (1968)



The IPA!

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	ɹ	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

IPA Bits and Pieces

Symbol

Diacritics

d^h_ɹ

IPA Bits and Pieces

You *really* should not assume that the un-decorated symbol is more “basic” or “general” or “simpler” than the decorated symbol.

diss (to insult)

[dɪs]

dis (this)

[d̪ɪs]

Place & Manner

Place

- Where the point of greatest constriction is.



Manner

- *What* happens at that place of constriction.

Place

Place

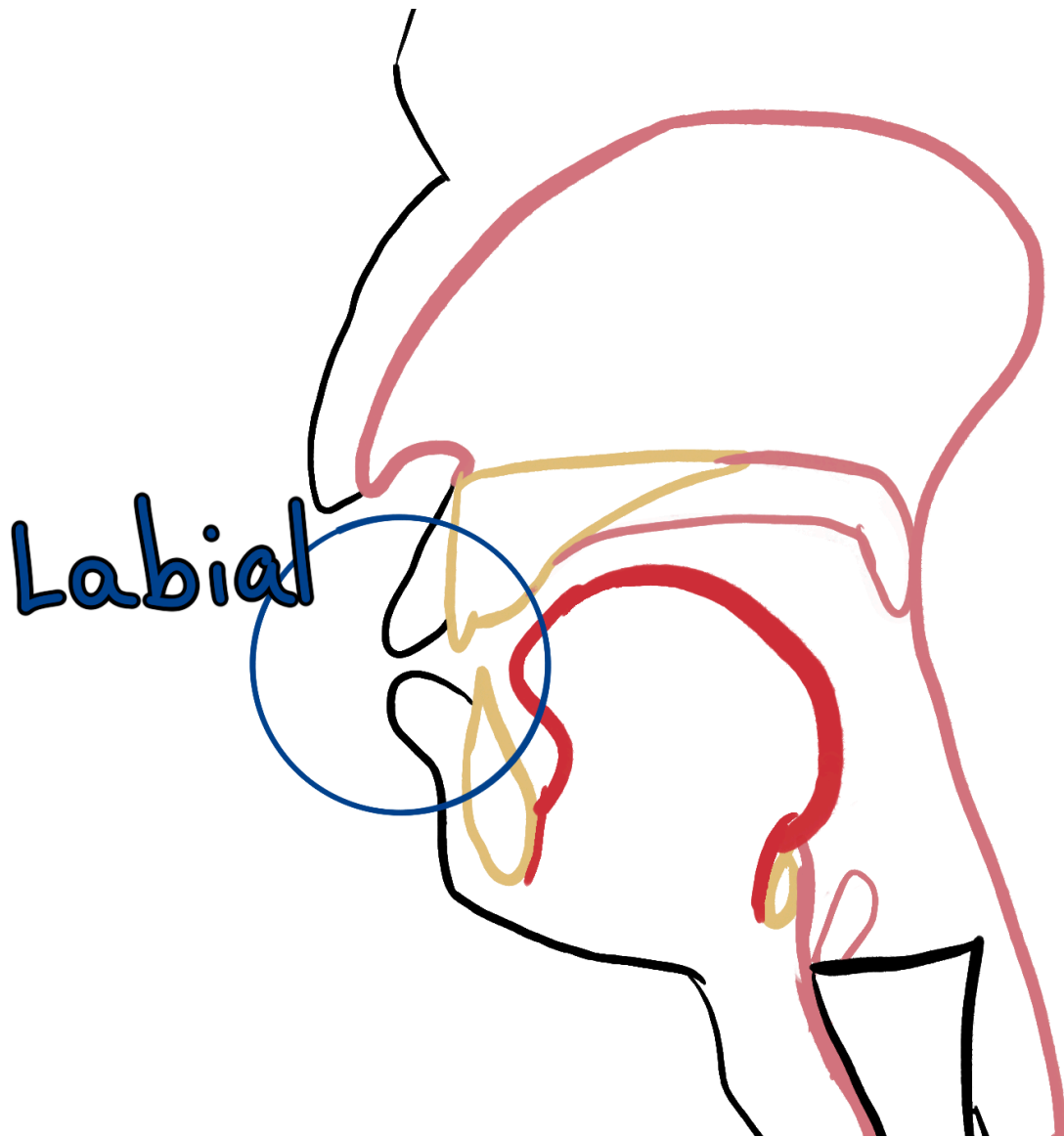
Most articulations have an *active* and a *passive* articulator

- Active: the articulator that moves
- Passive: the articulator that gets moved to

Most places of articulation are named for the location of the passive articulator.

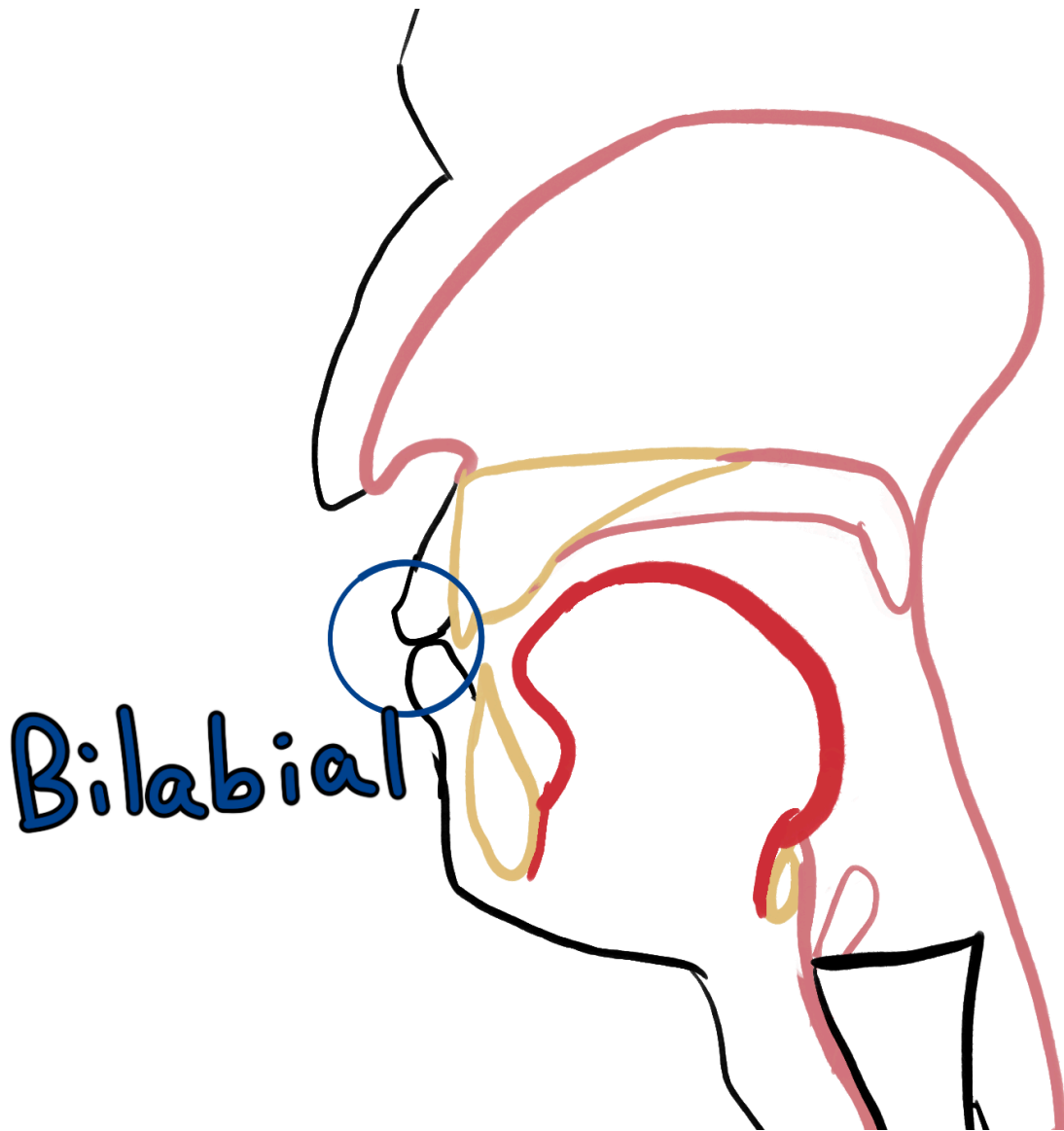


Place





Place



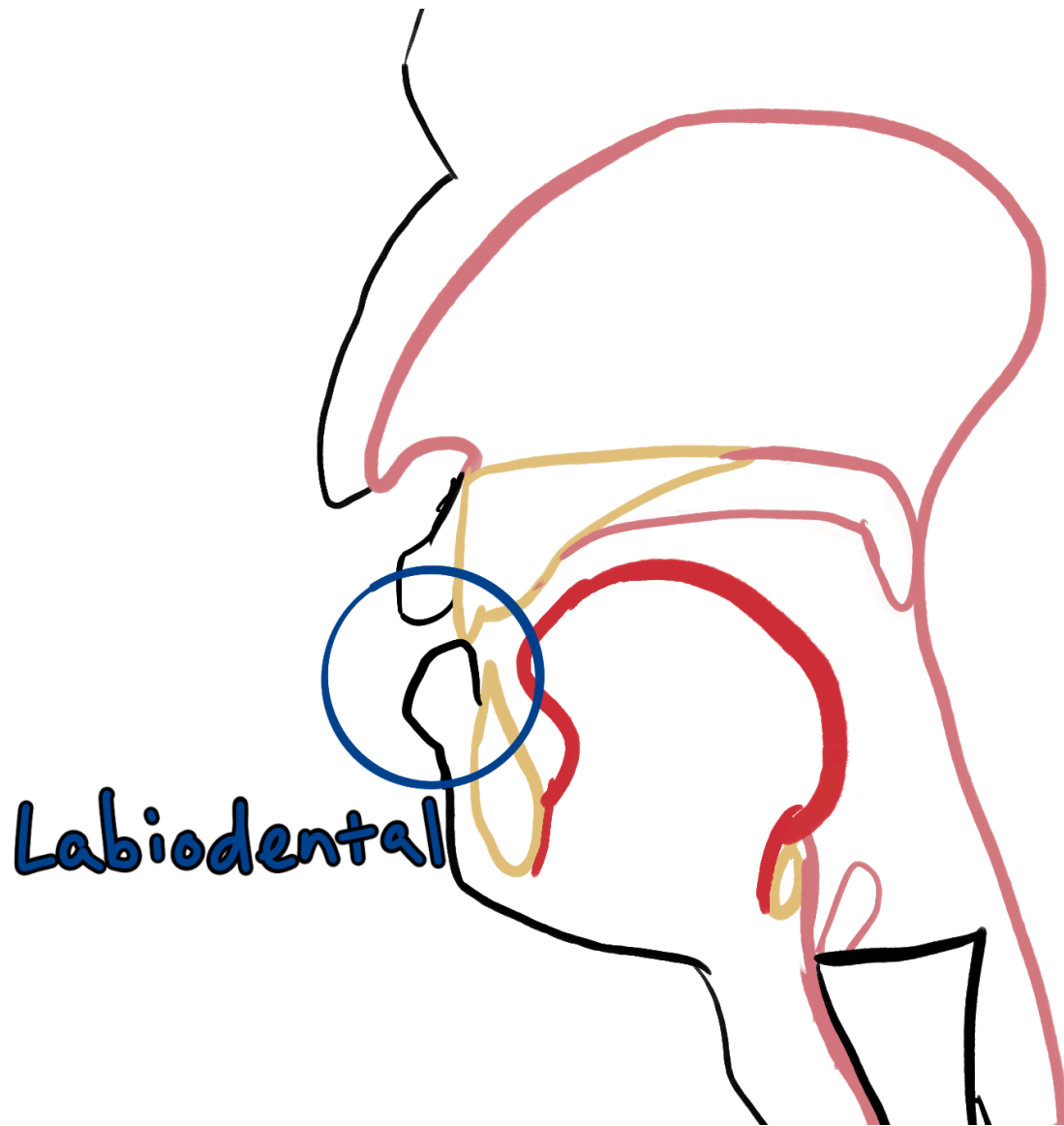


Place

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			



Place



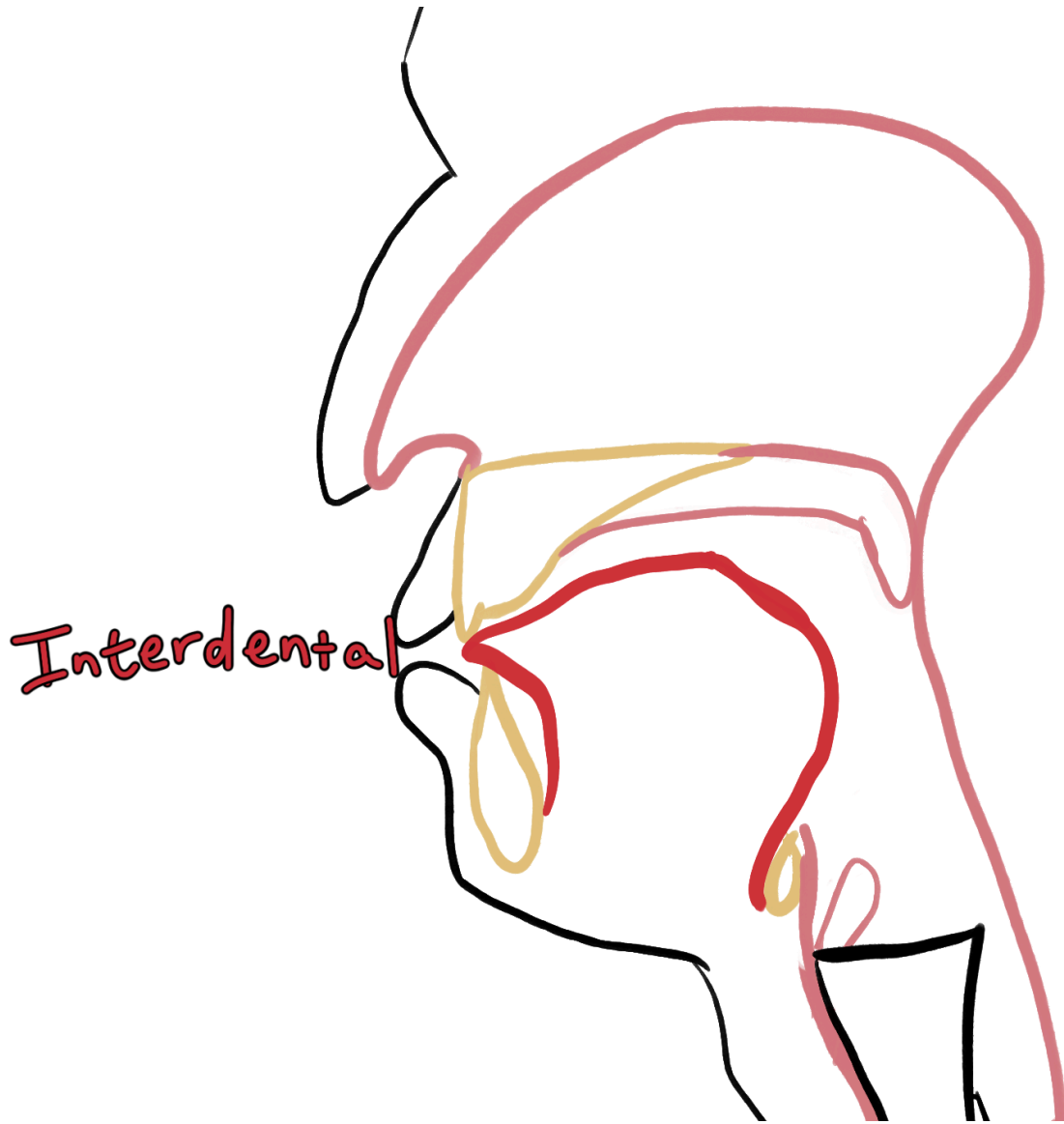


Place

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			



Place



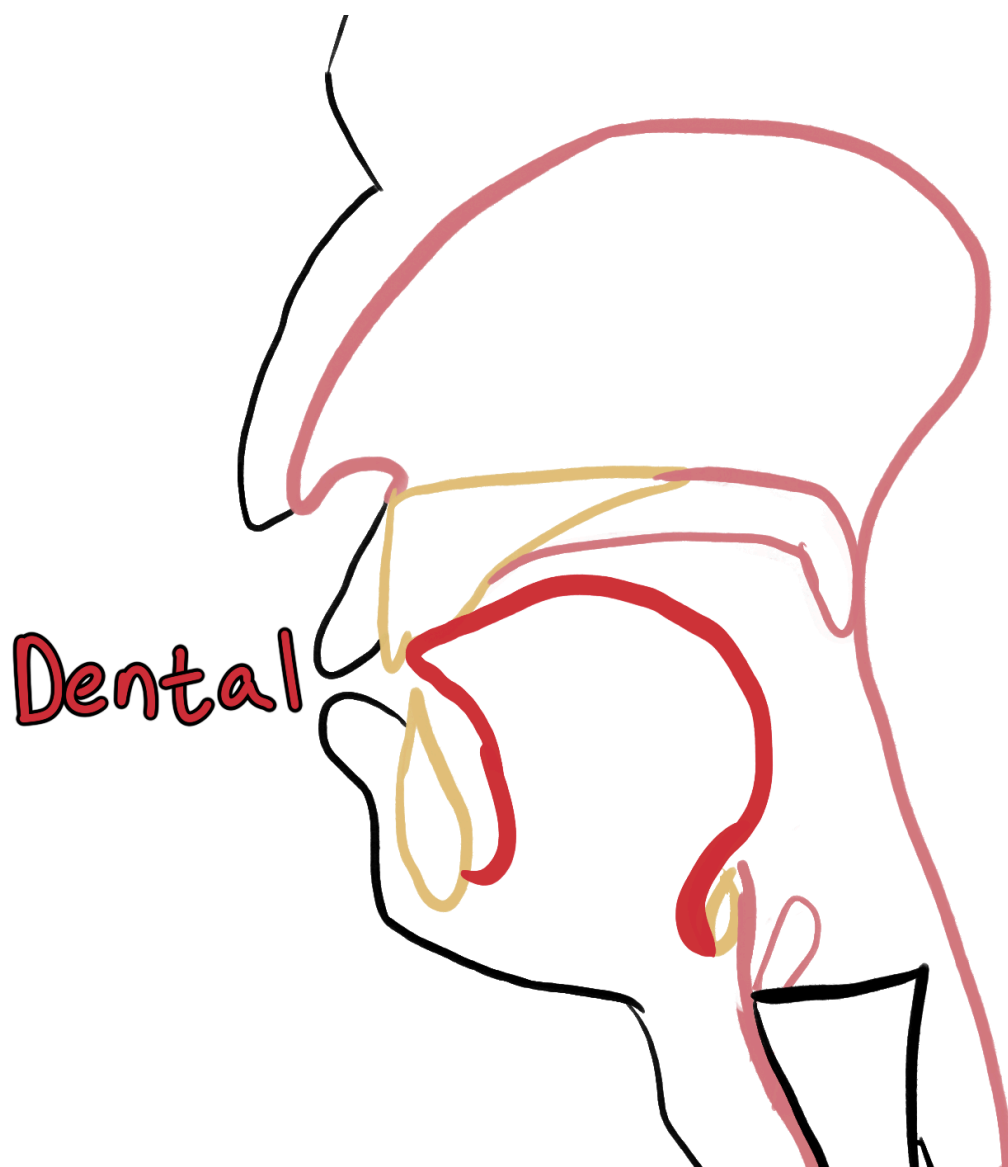


Place

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
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Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			



Place



Place

Dental

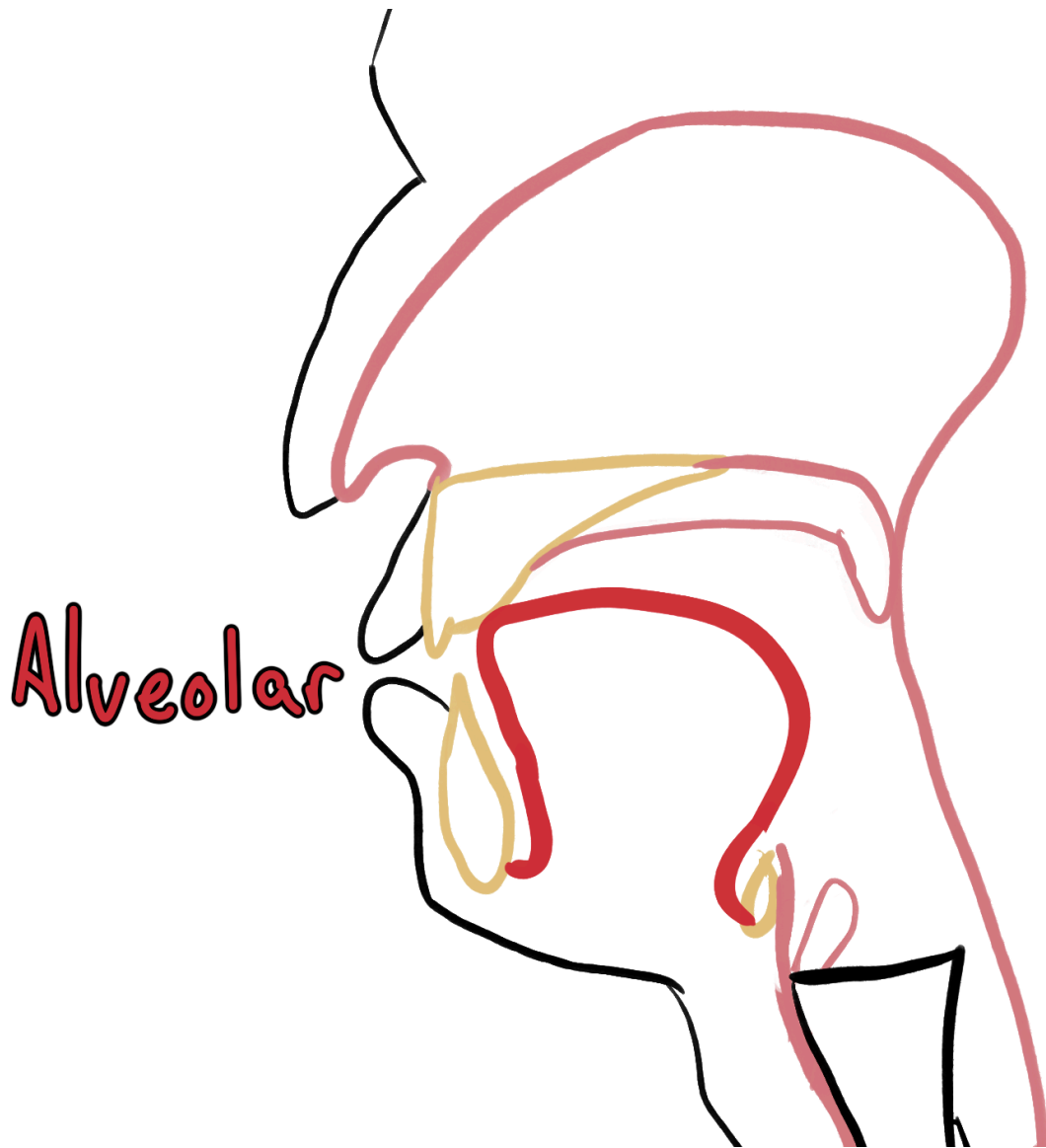
t d

n

s z



Place



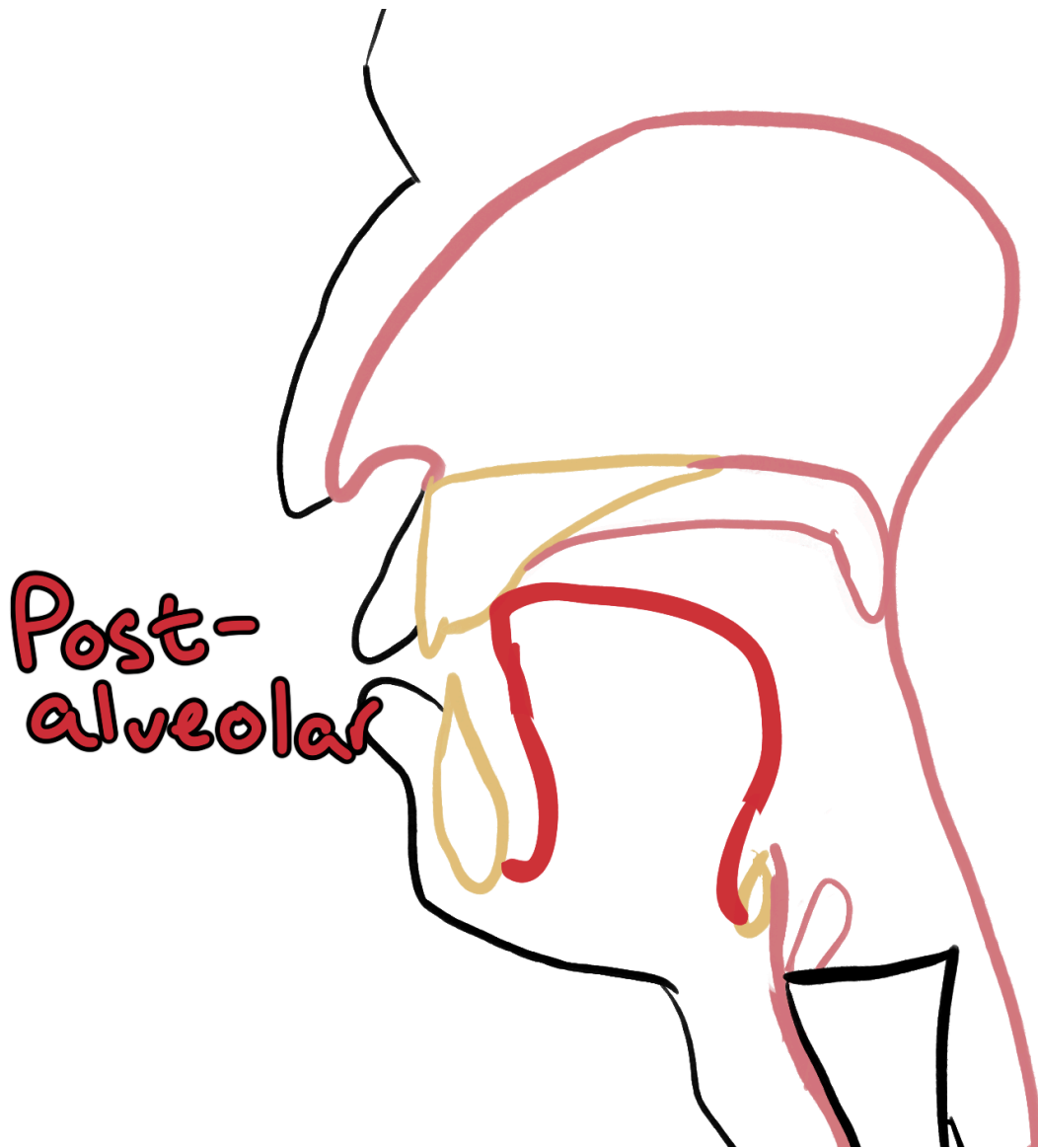


Place

	Bilabial		Labiodental		Dental		Alveolar		Postalveolar	Retroflex		Palatal		Velar		Uvular		Pharyngeal		Glottal	
Plosive	p	b					t	d		ʈ	ɖ	c	ɟ	k	g	q	ɢ			ʔ	
Nasal	m	ɱ					n			ɳ	ɲ		ɲ		ŋ	ɴ					
Trill	ʙ						r									ʀ					
Tap or Flap			v				ɾ			ɽ											
Fricative	ɸ	β	f	v	θ	ð	s	z	ʃ	ʒ	ʂ	ʐ	ç	ʝ	x	χ	ʁ	ħ	ʕ	h	ɦ
Lateral fricative							ɬ	ɮ													
Approximant	w	ʋ					ɹ			ɻ		j		ɰ							
Lateral approximant							l			ɭ		ʎ		ʟ							



Place



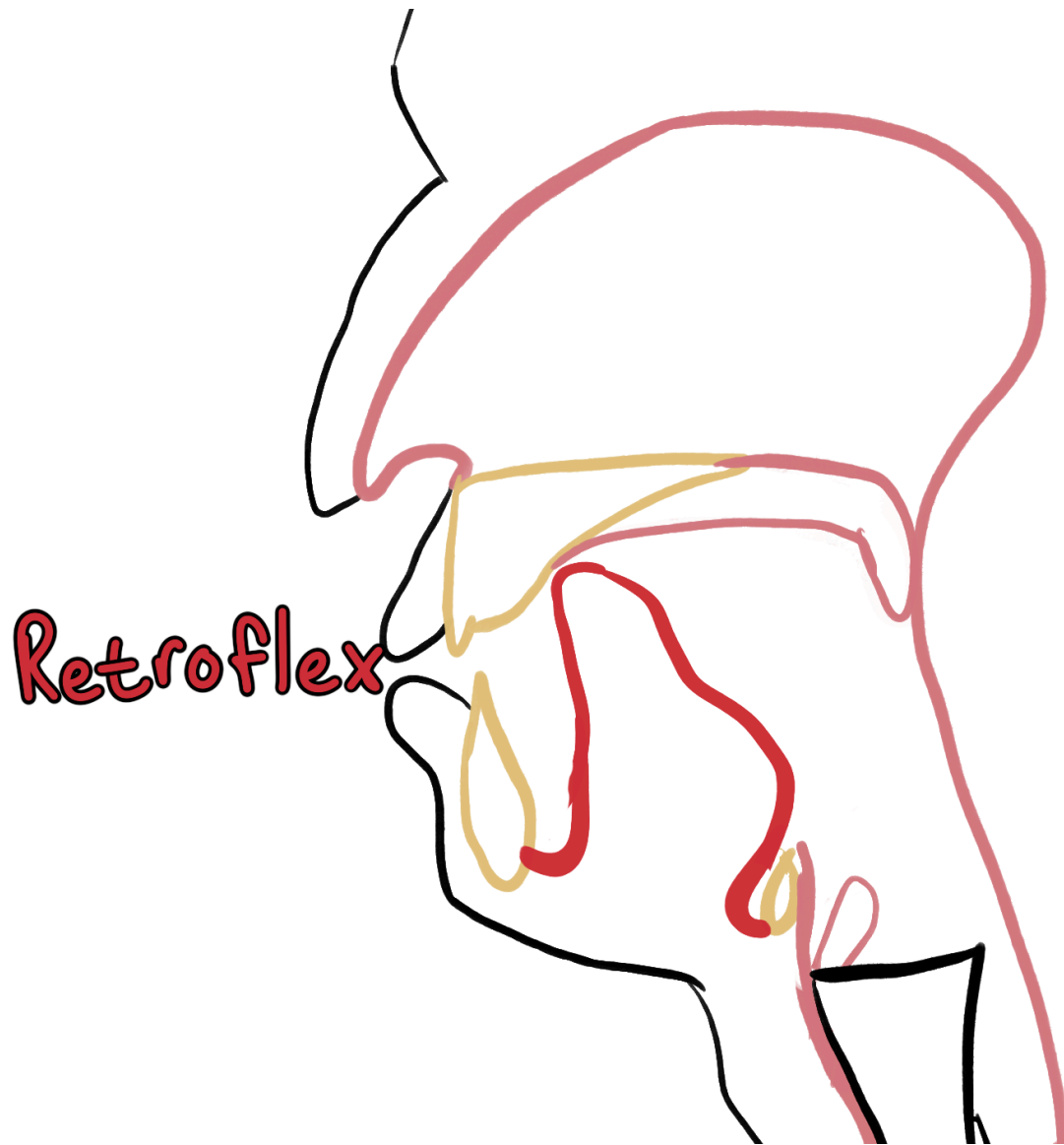


Place

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			



Place



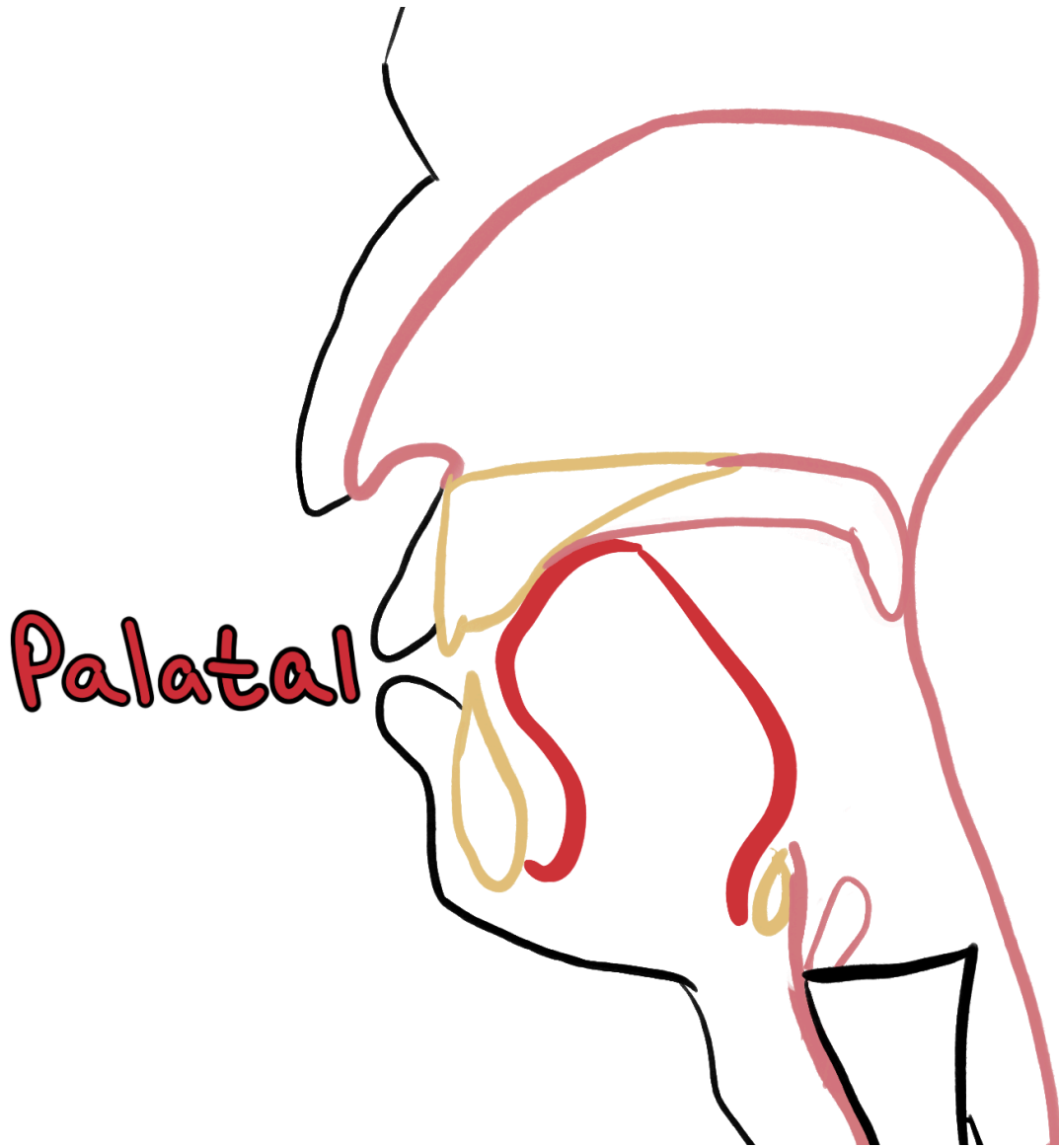


Place

	Bilabial		Labiodental		Dental		Alveolar		Postalveolar	Retroflex		Palatal		Velar		Uvular		Pharyngeal		Glottal	
Plosive	p	b					t	d		ʈ	ɖ	c	ɟ	k	g	q	ɢ			ʔ	
Nasal	m	ɱ					n			ɳ		ɲ		ŋ		ɴ					
Trill	ʙ						r									ʀ					
Tap or Flap			v				ɾ			ɽ											
Fricative	ɸ	β	f	v	θ	ð	s	z	ʃ	ʒ	ʂ	ʐ	ç	ʝ	x	χ	ʁ	ħ	ʕ	h	ɦ
Lateral fricative							ɬ	ɮ													
Approximant	w	ʋ					ɹ			ɻ		j		ɰ							
Lateral approximant							l			ɭ		ʎ		ʟ							



Place



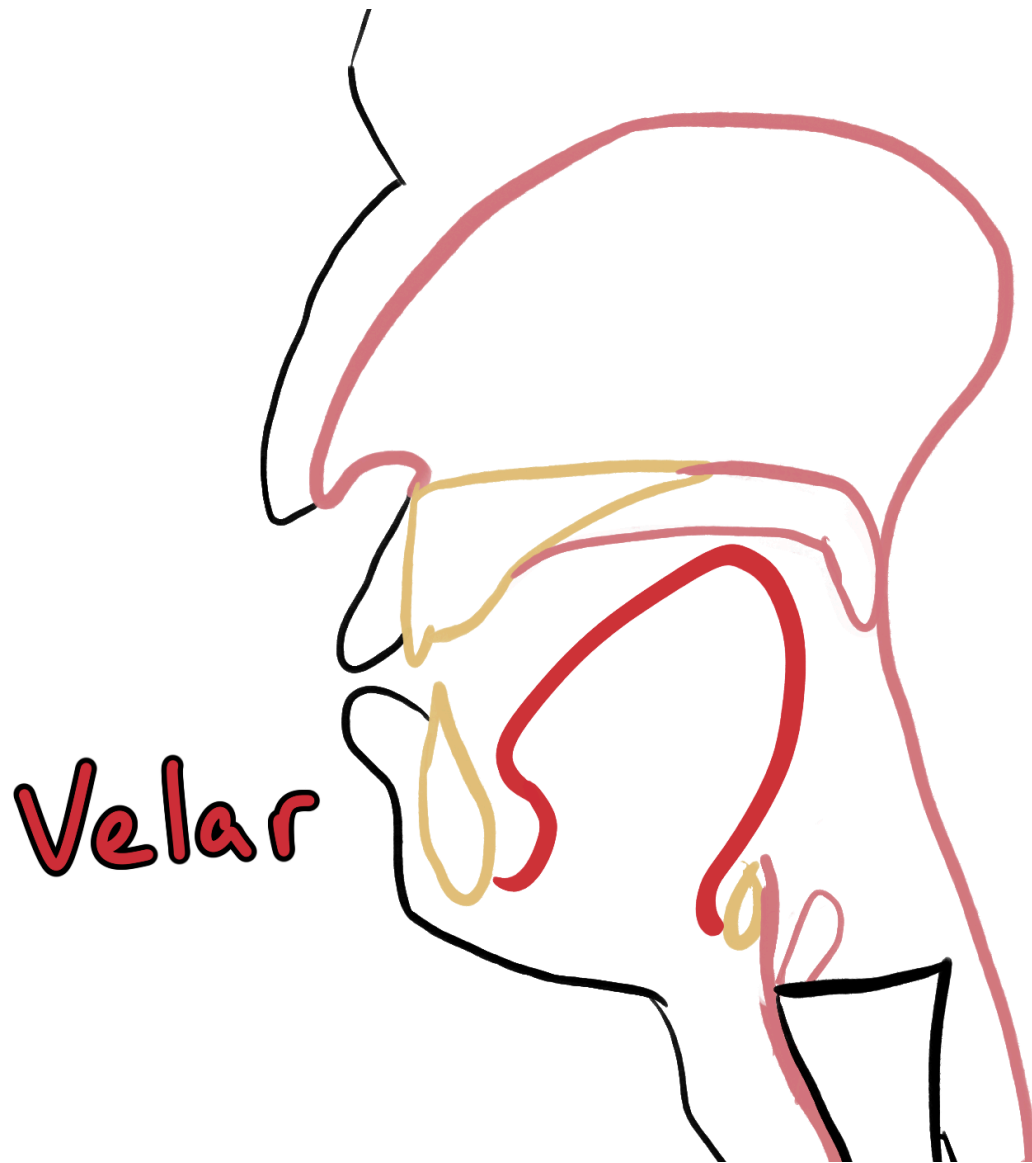


Place

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			



Place



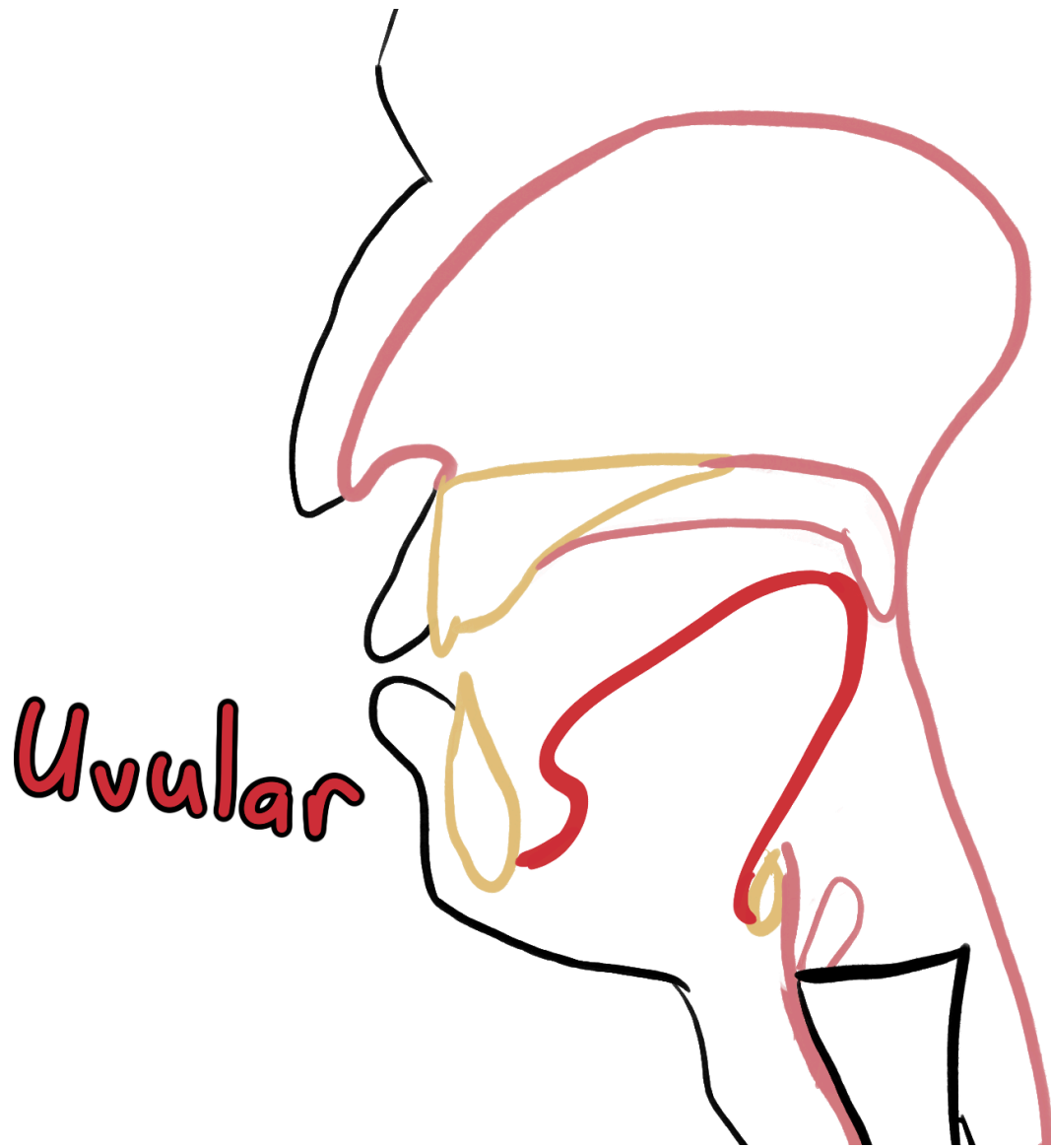


Place

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
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Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			



Place

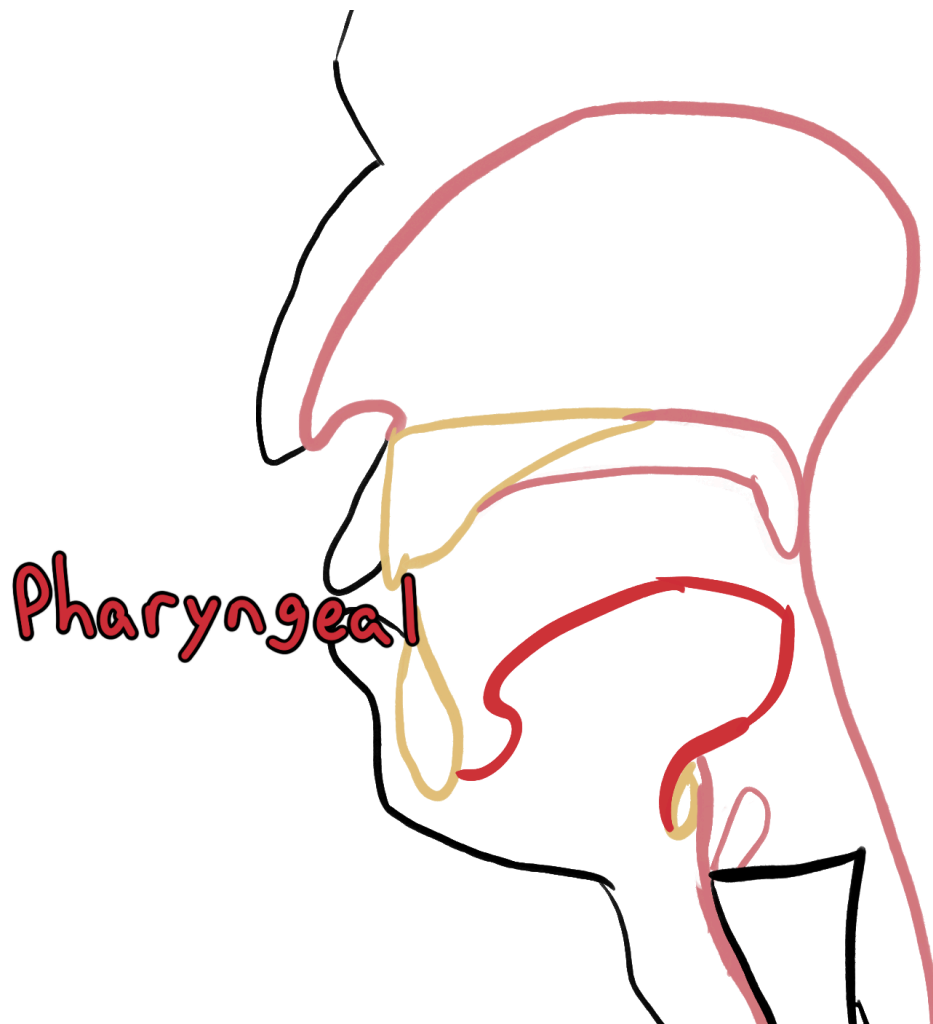




Place

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Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	ʋ	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Place





Place

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Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	ʋ	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Place

There are also terms based on the active articulator involved.

- Front of tongue: Coronal
 - Tip of tongue: Apical
 - Blade of tongue: Laminal
- Back of tongue: Dorsal

Place

Apical	Laminal
ʂ, ʐ	ç, ʝ
(retroflex)	(palatal)

Manner

Manner

What happens at each place of articulation.

Manner

Oral Stops/Plosives

- Full constriction between articulators
- No oral airflow
- No nasal airflow



Manner

Oral Stops/Plosives

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Manner

Nasal Stops

- Full constriction between articulators
- No oral airflow
- Nasal airflow

Manner

Nasal Stops

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			



Manner

Trills & Taps

- Trills
 - Articulators held loosely together, allowing for airflow to generate the Bernoulli effect
- Taps
 - A very fast, glancing articulation.

Manner

Trills & Taps

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			ʀ					ʁ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Manner

Nasals and trills are assumed to be voiced, but if they're voiceless we use an under-dot to indicate that.

ɱ ɱ̥
ʀ ʀ̥

Manner

Fricatives

- Close, but not complete constriction of articulators
- Turbulent airflow



Manner

Fricatives

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	ʋ	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Complex Consonants

Complex Consonants

- Doubly Articulated
- “Contour” segments



Complex Consonants

Doubly Articulated

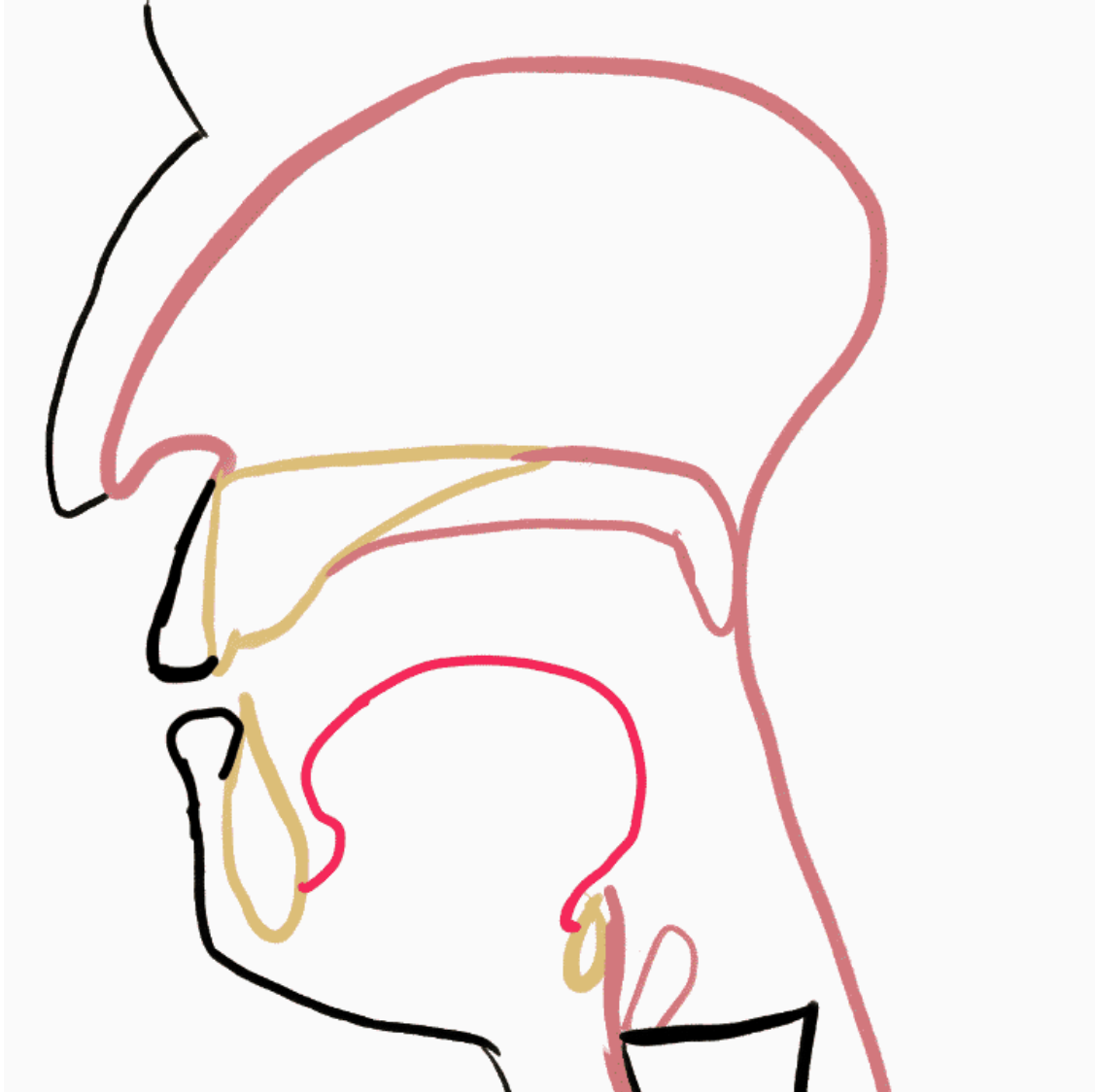


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Complex Consonants

Doubly Articulated

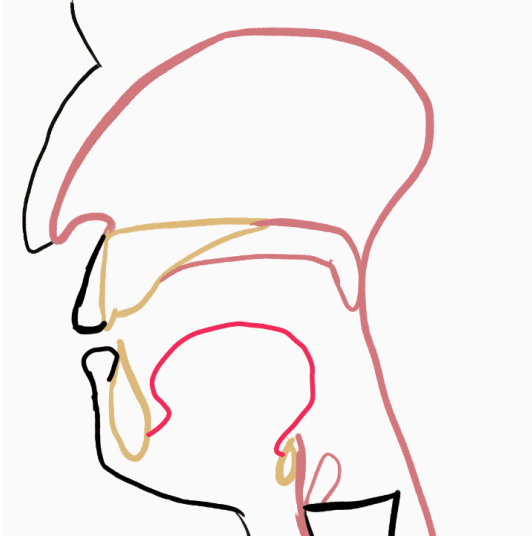


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Complex Consonants

Contour Consonants



$\widehat{d}_3$

Complex Consonants

Contour Consonants



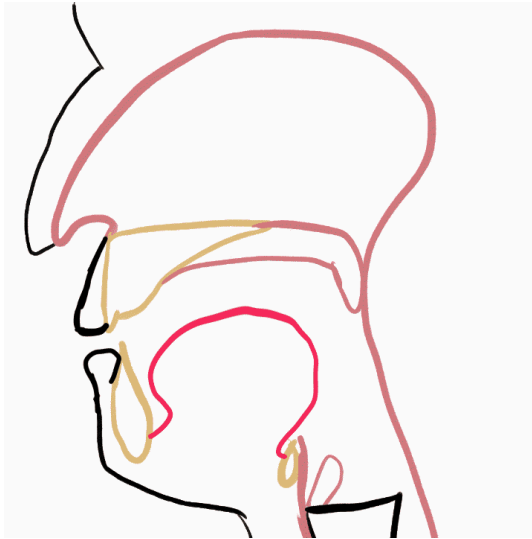
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Sproat & Fujimura (1993)



Complex consonants

Contour Consonants

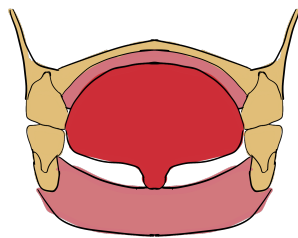


ɬ

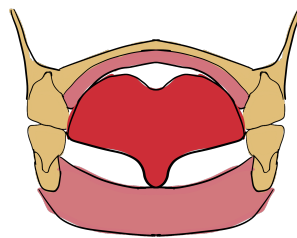
Sproat & Fujimura (1993)

Other Classifications

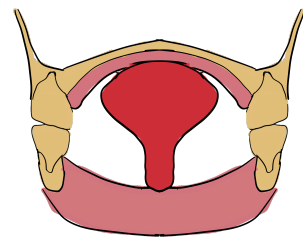
Laterals



Stop closure



Fricative constriction



Lateral constriction

“Rhotics”

Rhotics aren't a phonetically coherent set of sounds. Sounds classified as rhotics include

- alveolar trills
- uvular trills
- uvular fricatives



- velar fricatives
- alveolar taps
- retroflex approximants
- alveolar approximants
- whatever bunched [ɹ] is

“Rhotics”

As the name would suggest, these are sounds spelled with characters derived from Greek “ρ”.

References

Bibliography

Delattre, P., & Freeman, D. C. (1968). A dialect study of American r's by x-ray motion picture. *Linguistics*, 6(44). <https://doi.org/10.1515/ling.1968.6.44.29>

Sproat, R., & Fujimura, O. (1993). Allophonic variation in English /l/ and its implications for phonetic implementation. *Journal of Phonetics*, 21, 291–311.